ClueScope: Evidence-Centric Multimodal Video Retrieval for Interactive Search

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Abstract. In textual and conversational known-item search at the Video Browser Showdown (VBS), hints about a target are released over time, and the operator must decide which hint each candidate satisfies, and where, before submitting. Re-querying the accumulated description ranks well but discards this evidence, while autonomous agents take the decision from the operator. In ClueScope, visual, scene-text, speech and audio rankings are fused into video groups that keep frames, timestamps and channel provenance. Progressive Hint Memory (PHM) keeps an editable hint ledger and fuses per-hint rank evidence with the cumulative ranking, plus frontier-gated in-video searches. Beside the operator, our EDGAR agent controller runs DeepSeek-V4.1-Flash and Claude Code with an evidence board, typed verification and a stop guard; only the operator submits. A controlled study on 60 held-out synthetic three-hint queries over the AIC 2026 corpus (1.34 million keyframes, 873 videos) compares PHM with cumulative retrieval and independent-hint fusion with two encoders. With PE+Qwen, PHM's mean-prefix video MRR (0.6903) is close to cumulative retrieval (0.6907; 95% interval of the difference $[-0.046, 0.043]$), establishing neither superiority nor equivalence, and exceeds independent-hint fusion (0.6662), at 3.88 versus 1.13 s median latency. Cumulative retrieval thus stays the default ranking, and PHM an opt-in per-hint evidence view.

Keywords: Interactive video retrieval · Multimodal search · Progressive hints · Evidence inspection

1 Introduction

As hints accumulate in textual and chat Video Browser Showdown (VBS) tasks [16], operators must inspect which hints support each candidate and where. A *hint* is a released description increment, a *query* a retrieval request, a *channel*

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a searchable modality/model, a *candidate* a proposed video or moment, *evidence* a source-linked observation, and a *verdict* its support for a hint.

VBS 2027 emphasizes textual/chat known-item search (KIS-T/KIS-C), VQA and ad-hoc video search (AVS), alongside visual KIS [20]. ClueScope contributes:

- **Multimodal evidence integration.** Visual, OCR, speech and audio rankings and optional multi-scale clips keep timestamps and provenance.
- **Inspectable progressive search.** PHM links an editable hint ledger to per-hint rank evidence, with a controlled study against cumulative retrieval.
- **Verification workflow with agents.** Spatial queries, VQA, an AVS basket and the EDGAR controller support operator-reviewed DRES submission.

2 Related Work

PraK V4 supports spatial and within-video search [10]; NII-UIT adds VQA [19], Exquisitor conversational and temporal search [11,18], and SnapMind editable plans [8]. ClueScope emphasizes persistent hint-specific evidence and local-match ranking eligibility, with operator-controlled submission rather than autonomous retrieval [9]. Our EDGAR system [15] combines two agents and typed verification; ClueScope reuses its controller with DeepSeek in Codex's role, and its AIC results evaluate neither PHM nor this configuration.

3 Multimodal Retrieval Architecture

Indexed evidence. The AIC 2026 collection [6] has 873 videos and 1,339,055 keyframes, using InfoShot-style sampling [22] extended for short events and text changes. Milvus [21] searches separate normalized PE-Core-G14-448 [2] (1,280-d) and Qwen3-VL-Embedding-8B [13] (4,096-d) vectors. Elasticsearch holds metadata, OCR, speech, audio and objects.

Query interpretation and fusion. Heuristics or an optional LLM extract per-channel queries, translating visual descriptions while preserving original-language OCR/speech. For frame f, weighted reciprocal rank fusion (RRF) [3] computes

$$R(f) = \sum_{c: f \in L_c} \frac{w_c}{60 + r_c(f)}, \qquad r_c(f) \geq 1, \tag{1}$$

where L_c is channel c's frame ranking, $r_c(f)$ its one-based rank, w_c a preset weight (1.0 visual, 0.8 OCR and speech, 0.7 audio) and 60 the RRF constant, so raw scores of different embedding spaces are never added.

Video grouping. For video v with frames F_v and best frame $f_v^\star$,

$$S(v) = 0.75\frac{R(f_v^\star)}{R^\star} + 0.15\frac{\mathrm{mean}_{\mathrm{top3}(F_v)}\, R}{R^\star} + 0.10\min\Big(1, \frac{n_{30}(v) - 1}{4}\Big), \tag{2}$$

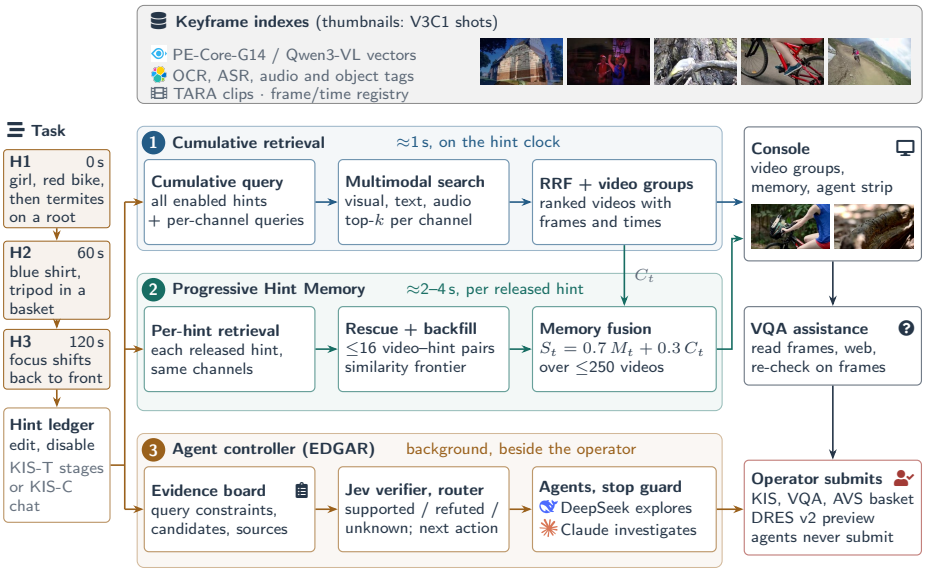


Fig. 1. ClueScope: editable hints drive cumulative retrieval (1), PHM fusing per-hint evidence with cumulative ranking C_t (2), and the EDGAR controller (3); only the operator submits. PHM consumes pre-TARA groups. Latencies: Table 1 medians.

where $R^\star$ is the top frame score over all videos, the mean covers at most three frames, $n_{30}(v)$ counts frames within 30 s of $f_v^\star$ (itself included), and ties break by video id. These fixed presets let the strongest frame dominate; they were not tuned or ablated. Optional TARA [1] indexes 168,536 clips over 8-, 24- and 72-s windows; each scale gives at most one vote per video, and RRF fuses scale and frame-channel video rankings.

4 Interactive Search and Evidence Verification

Visual KIS via spatial queries. Operators place objects, colors, positions and actions on a canvas; Hungarian assignment [12] matches them to detections and RRF adds PE searches of generated captions and any low-weight sketch.

Evidence-supported VQA. Up to 12 keyframes (the selected one and up to three per top group, at least 2 s apart) are labelled C01, C02, ... with their OCR, speech and audio text; DeepSeek-V4.1-Flash [5] or NVILA [14] marks those showing the event and proposes answers tied to them. If no answer is visible or the model-reported, uncalibrated confidence is below a preset 0.55, a web-grounded model proposes answers that are re-checked on the same frames; contradicted answers are listed as rejected with the visible reason, and conflicts are shown with their cited frames and sources. NVILA latency is unmeasured.

Agent controller. ClueScope runs the EDGAR controller [15] beside the operator, with DeepSeek-V4.1-Flash 1017 [4] in the exploring role Codex held there and Claude Code (Opus 5.5) as investigator. Each applied description gets a fresh evidence board of query constraints, retrieved candidates and observation sources; an agent's report counts as visual evidence only after it opened the frame. Jev 1.13 judges the leading candidates as supported, refuted or unknown and picks the next action: stop, or call DeepSeek, Claude or both. A stop guard accepts a stop only if the top candidate's raw, uncalibrated Jev probability reaches 0.9, leads competitors by 0.1 and every applicable constraint is judged supported; these EDGAR presets make stopping a policy decision, not a correctness guarantee. Otherwise the next agent runs, within six steps, 40 tool calls and 240 s. DeepSeek runs at high reasoning as a backend tool loop. This configuration has not been re-run on the EDGAR benchmark; a three-turn smoke test (5.3 s, $0.0006) shows only that the loop runs. Proposals appear in the agent strip; agents never submit, and a new hint restarts the controller on the cumulative description.

AVS and DRES submission. An *anchors-only* queue applies 2-s temporal non-maximum suppression and proposes one anchor per video per pass; operators verify playback and collect moments in a task basket blocking exact duplicates. The DRES [17] v2 adapter supports login, task/hint discovery, millisecond coordinates, preview and verdict history. Operators confirm every submission and a changed task blocks sending; mock-transport tests cover AVS routing, temporal payloads and duplicate protection, but live-server tests and log delivery are pending.

5 Progressive Hint Memory

The evaluated PHM variant keeps released, enabled hints in an editable ledger (Fig. 2). Duplicates add no vote; cumulative appends contribute their new suffix; edits invalidate superseded evidence. Global lists contain only top-k frames ($k = 200$ per model/view in the study). Bounded within-video *rescue* checks survivors against new hints; *backfill* checks newcomers against earlier hints.

After per-hint fusion and grouping (Eqs. 1–2), the rank $r_i(v)$ of video v for hint i gives evidence $e_i(v) = 61/(60 + r_i(v))$; $C_t(v)$ applies the same transform to v's current cumulative rank (0 if absent). I_t, shared by all candidates, holds the hints whose retrieval had a healthy channel: a failed hint is left out, while a valid hint that misses v gives $e_i(v) = 0$ and an unobserved label, as truncation does not imply irrelevance. Over the previous, cumulative and latest-hint candidates,

$$M_t(v) = \left[\prod_{i \in I_t} \big(0.10 + 0.90 e_i(v)\big)\right]^{1/|I_t|}, \qquad S_t(v) = 0.70 M_t(v) + 0.30 C_t(v). \quad (3)$$

These fixed heuristic presets are not probabilities and are not individually ablated: 0.10 floors missing rank evidence and 0.70 favours memory over the current cumulative query, so early-hint distractors can persist. Later turns search up to eight survivors lacking new-hint evidence and backfill older hints for up to four

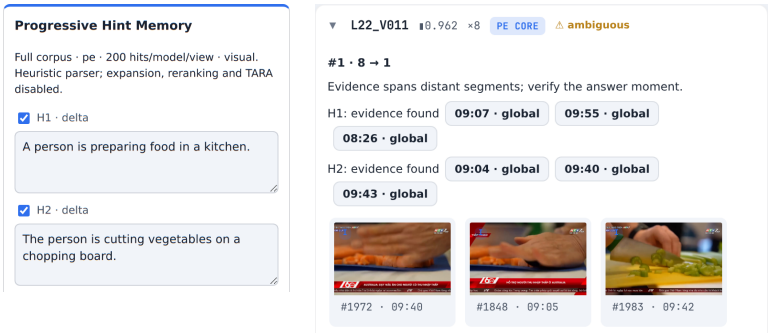


Fig. 2. PHM: released hints (left); candidate history, evidence origins, and frames (right). This archived PE-only session predates the benchmark's frontier revision.

newcomers ($\leq$16 video–hint pairs, three frames per model/pair). Local hits rank only if their raw similarity reaches the lowest finite similarity in the original global list for the *same model, query, and collection* (the *similarity frontier*); eligible hits are deduplicated and re-ranked with that list (candidate-pool, not corpus, ranks), while hits below the frontier, or lacking one, remain localization-only evidence. Memory retains at most 250 videos, protecting current/previous top-20 candidates. Given the results, the console's default ranking stays cumulative and PHM (0.70 preset) is an opt-in KIS-T/KIS-C view; its per-hint labels and rank history are retrieval evidence to inspect, not proof a hint holds.

6 Controlled Component Evaluation

The 75 synthetic queries come from our AIC 2026 practice set. Around each legacy ground-truth frame, Codex inspected contact sheets decoded from the source video (median one frame per second; not the index keyframes), drew half-open answer intervals at that resolution (one target video per query; 88 intervals, median 6 s) and split the description into three hints; a co-author accepted both unchanged. Fifty hint sets span a multi-shot context and 25 a single moment; no independent annotation or playback check was made. Target-video components separate 15 development and 60 held-out queries (55 groups), each with three prefixes. PE and PE+Qwen search all 1,339,055 frames at depth 200 per model/view, with code (`b012b9b`), annotations and translated parser plans frozen and auxiliary modalities, TARA, expansion and reranking off. Baselines are cumulative retrieval of all released hints, Hint-RRF (an equal-weight sum of $e_i(v)$ over I_t across the full observed union, without memory or rescue) and PHM without local searches. Video MRR, capped at 250 videos, averages prefixes then queries; latency is the median wall time per prefix of the whole method, observed in one run per profile, executed sequentially on live remote encoder and Milvus services with session caches and no warm-up; hardware and load were not controlled. The 1,800 held-out method prefixes (180 each: six PE methods, including untabulated

Table 1. Mean-prefix video MRR and median latency on 60 held-out queries.

	PE		PE+Qwen	
Method	MRR	p50 (s)	MRR	p50 (s)
Cumulative	0.6558	0.50	0.6907	1.13
Hint-RRF	0.6093	0.51	0.6662	1.11
PHM without rescue	0.6230	0.83	0.6832	1.52
PHM	0.6459	2.13	0.6903	3.88

dual-view and depth-400 cumulative controls, and four PE+Qwen ones) passed prefix-isolation and metric audits (hint isolation, not annotation quality).

Cumulative retrieval is a strong ranking baseline (Table 1): PHM differs by -0.0099 for PE (95% interval $[-0.0576, 0.0358]$) and -0.0004 for PE+Qwen ($[-0.0460, 0.0431]$), establishing neither superiority nor equivalence. It exceeds Hint-RRF by 0.0367 and 0.0241 with positive unadjusted intervals (10,000 paired video-group bootstrap resamples [7]); comparisons are exploratory. PHM gains late and loses early: with PE+Qwen it ranks the target first after the third hint more often (73% vs. 68%) but after the second less often (57% vs. 63%); with PE it beats cumulative retrieval on 35 prefixes and loses on 21, a loss costing twice the reciprocal rank of a win (0.52 vs. 0.26). Only 3 of 180 translated PE inputs, one per query, exceed PE's 70-token window, by 1–3 tokens; excluding them post hoc (57 queries remain), PHM minus cumulative is $+0.0031$ for PE ($[-0.0396, 0.0452]$) and $+0.0025$ for PE+Qwen ($[-0.0445, 0.0466]$). Rescue recovered no target beyond the 250-video cap and worsened 20 PE and 10 PE+Qwen prefixes. Video MRR measures shortlisting only. Final-hint moment Hit@1 (the top group's first displayed frame in an answer interval) is 24/60 for PHM and cumulative retrieval alike (PE+Qwen), a tie: each hits four queries the other misses. PHM reuses the cumulative view's frame within each video, so where both rank the target video first (35 queries) the outcomes coincide; PHM's extra top-video hits (44 vs. 41) add no moment hits. Findings may not transfer to V3C.

7 Conclusion

On AIC 2026, cumulative retrieval remains strong and PHM adds per-hint evidence at extra latency. Future work will examine PHM on VBS data, DeepSeek agents, operator effort, and whether memory and agents reduce wrong submissions.

Acknowledgments. We thank the VBS and AIC 2026 organisers and the reviewers. This research is partially supported by the research funding from the Faculty of Information Technology, University of Science, Ho Chi Minh City, Vietnam.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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